0580_m22_ms_22_Q21

Question	Answer	Marks	Partial Marks
21(a)	35.1 or 35.05 to 35.06	**4**	**M3** for $\tan = \frac{14.5}{\sqrt{18.6^2 + 9^2}}$ oe or **M2** for $\left[AC^2 =\right] 18.6^2 + 9^2$ oe or better or $\left[AG^2 =\right] 18.6^2 + 9^2 + 14.5^2$ or **M1** for recognising the angle *GAC*
21(b)	$30 - \sqrt{18.6^2 + 9^2 + 14.5^2}$ $30 - \frac{14.5}{\sin(their\,\mathbf{(a)})}$ or $30 - \frac{\sqrt{18.6^2 + 9^2}}{\cos(their\,\mathbf{(a)})}$	**M2**	**M1** for $AG^2 = 18.6^2 + 9^2 + 14.5^2$ oe or better or $\sin(their\,\mathbf{(a)}) = \frac{14.5}{\text{AG}}$ or $\cos(their\,\mathbf{(a)}) = \frac{\sqrt{18.6^2 + 9^2}}{\text{AG}}$
	4.75 to 4.78…	**A1**	